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Anomaly Detection Using Generative Language Models and Deep Feature-based Time Series Similarity

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ABSTRACT Time series data play a critical role in decision-making across domains such as environmental monitoring, industrial equipment management, and financial market analysis. However, these data are highly susceptible to distortions caused by noise, anomalies, and sensor failures. Early detection and interpretability of abnormal patterns are essential to ensuring the reliability of data-driven systems. This paper proposes a novel anomaly detection framework that integrates large language models (LLMs) with time series retrieval-augmented generation. Traditional anomaly detection methods often rely on statistical models or single deep learning architectures, which struggle to capture irregular patterns and lack interpretability. Our method segments time series into interpretable subsequences, embeds them into high-dimensional vectors, and retrieves similar historical cases from a vector database. In addition, structural similarity across spatially adjacent sensors is quantified using Dynamic Time Warping (DTW) and Z-score metrics. These analytical results are then used to construct prompts, which are passed to an LLM to generate natural language decisions and explanations. The proposed framework enables both accurate detection and human-readable reasoning. Evaluated on real-world air quality monitoring data from South Korea, our approach reduced the false positive rate by over 50%, while maintaining recall, thereby validating its practical reliability and explainability compared to a segmentation-only approach.

INDEX TERMS Anomaly detection, Large Language Model, Retrieval-augmented generation, Time series, Time series segmentation

I. INTRODUCTION

TIME series data play a critical role in various real-world applications, including environmental monitoring, industrial system diagnostics, and financial forecasting. However, such data are often prone to noise, outliers, and sensor errors, making early anomaly detection and interpretation vital for reliable decision-making.

Traditional anomaly detection relies on statistical models or single-stream deep architectures, which often miss complex, irregular patterns and offer limited support for identifying spatially correlated anomalies or producing human-interpretable outputs [1], [2], [3], [4]. Moreover, in many cases, anomaly detection still depends on expert judgment, highlighting the growing demand for automated and consistent decision support systems. Concurrently, work on intrinsic predictability and signal propagation motivates context-aware anomaly detection: wavelet-entropy quantifies how predictable a series is versus noise, while complex-network studies show topology and nonlinear interactions drive spatio-temporal coupling [5], [6]. These results support incorporating segment-level predictability and spatial context into the pipeline. To address the interpretability challenge, one promising direction is to leverage natural language explanations alongside anomaly flags. Recent advances in large language models (LLMs) enable natural-language rationales for decisions [7], [8]. However, LLMs inherently lack structural awareness of time series data, particularly in capturing local temporal dynamics. Retrieval-augmented generation (RAG) has begun to bridge this gap in time series forecasting and reasoning [9], [10], [11].

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This study introduces a framework that combines retrieval-based contextualization and LLM-driven explanations for time series anomaly detection. The framework decomposes inputs into segments, embeds them in a high-dimensional space, and retrieves similar historical patterns from a vector database. Spatial comparisons help distinguish true anomalies from sensor-specific deviations. The aggregated evidence is then converted into a structured prompt for LLM-based, human-readable decisions and explanations.

Our evaluation on real-world air-quality data from South Korea [12], [13] shows that the proposed method reduces total alarms by 52.9% and false positives by 54.9% relative to a segmentation-only baseline, while preserving recall (88 vs. 89 true positives) and improving overall accuracy across five pollutants (CO, NO, NO_2, O_3, SO_2). These results indicate improved reliability without sacrificing sensitivity, suggesting applicability to broader time series domains.

The key contributions of this work are summarized as follows:

- *Segmentation for candidate anomalies:* A 1D U-Net based segmenter is employed to identify structurally abnormal subsequences in time series data.
- *Retrieval and spatial context:* We utilize T-Rep embeddings and DTW/z-score comparisons across neighboring sensors.
- *LLM-based explanations:* We build structured prompts that fuse retrieved evidence and spatial cues.

II. RELATED WORKS

A. TIME SERIES SEGMENTATION AND CHANGEPOINT DETECTION

Time series frequently undergo distributional or structural shifts. Detecting such changepoints enables segmentation into locally homogeneous subsequences and serves as a standard precursor to anomaly analysis

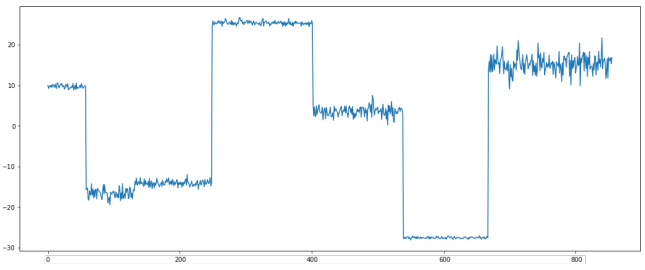


FIGURE 1. Synthetic time series with multiple abrupt mean shifts under additive noise. Although the level shifts are visible, estimating the exact change-point locations requires algorithmic localization due to noise and sampling effects.

Fig. 1 presents a synthetic time series with abrupt changes in mean values. Even in such clear-cut cases, accurately detecting the timing of change points can be challenging. Various change point detection algorithms have been proposed to automate this process.

Time series segmentation is a critical preprocessing step for anomaly detection, as it identifies structural changes within temporal data streams. Classical methods include Bayesian Online Change Point Detection (BOCD) [1], Binary Segmentation [3], and Pruned Exact Linear Time (PELT) [4], trading off optimality, speed, and robustness; however, their piecewise-stationarity assumptions often degrade under noise and nonstationarity

However, these traditional methods operate under strong statistical assumptions (e.g., piecewise stationarity or distribution shifts) and struggle with noisy or non-stationary real-world data. More importantly, they are limited in capturing complex temporal dependencies or higher-level structural features in the data. Deep encoders alleviate these limits: 1D U-Net [14] localizes boundaries via encoder–decoder paths and skip connections, while recurrent models [15] capture long-range dependencies but are generally less precise at boundary localization. We therefore adopt a 1D U-Net to propose candidate abnormal segments. This model forms the first stage of our anomaly detection pipeline, enabling robust detection of structurally abnormal subsequences even in the presence of noise or irregular sampling.

B. REPRESENTATION LEARNING FOR TIME SERIES

Self-supervised encoders learn transferable features without labels—valuable when anomalies are rare—by pretext tasks such as masking, ordering, and segment permutation [16], [17], [18], [19]. Effective representations must compress while preserving rare, abrupt deviations crucial for anomaly reasoning.

Our framework builds on T-Rep, a time-aware extension of the popular TS2Vec architecture as our primary encoder in this study [20], [21]. For context, TS2Vec [20] is a self-supervised framework that learns time-step representations by combining multi-resolution context with contrastive learning. As illustrated in Fig. 2, the architecture processes raw time series using dilated convolutions and random masking, followed by hierarchical contrastive objectives across both temporal and instance levels.

TS2Vec uses temporal and instance-level contrastive objectives to enforce representation consistency across time-steps and samples.

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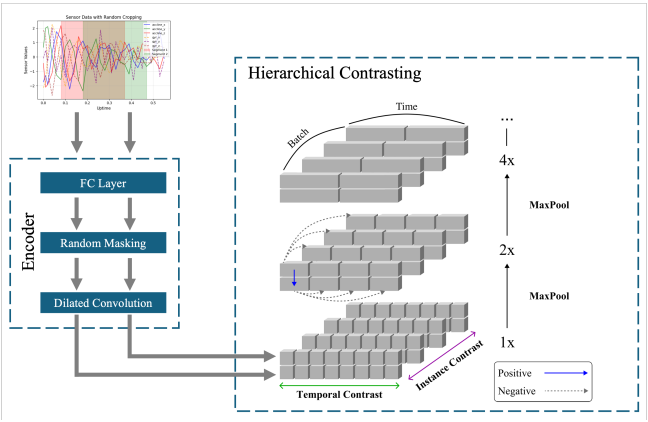


FIGURE 2. TS2Vec architecture: a 1-D dilated-convolution encoder with random masking produces embeddings, while hierarchical contrastive losses across time and instances with multi-resolution pooling pull positives together and push negatives apart to yield robust time-step representations.

Building upon these ideas, T-Rep integrates time-aware embeddings using Time2Vec [22] to enhance the encoder's sensitivity to position-specific structural changes. Additionally, it integrates divergence-based objectives and time-conditioned forecasting to improve robustness against noisy and missing values.

These enhancements enable T-Rep to learn robust, temporally structured representations. In our framework, these embeddings are stored in a vector database for similarity-based retrieval, forming the basis for contextually grounded natural language prompts. This ensures that anomaly decisions are not only accurate but also interpretable via retrieved evidence.

C. LARGE LANGUAGE MODELS FOR EXPLANATIONS

Based on the Transformer architecture [23] LLMs offer a promising way to generate human-readable anomaly explanations based on structured evidence. Unlike traditional models such as Word2Vec and Seq2Seq architectures, modern LLMs are pretrained on massive corpora and capable of zero-shot or few-shot generalization [7], [8], [24], [25], [26], [27].

Among these, the LLaMA series developed by Meta AI [8] has shown strong performance while maintaining computational efficiency. In this study, we utilize the LLaMA 3 family model [28] as the generative component in our anomaly detection pipeline. In particular, its multilingual instruction-following capability and 8B size enable effective and fast inference on commodity hardware.

Specifically, we design prompts that incorporate time series metadata, retrieved historical segments, and anomaly scores. The LLM then generates both a binary decision and an explanatory sentence in natural language. The prompts include both raw statistics and similarity-based anomaly evidence, enabling the LLM to reason based on concrete temporal context. This enables interpretability and user engagement, transforming raw anomaly detection outputs into human-understandable insights.

D. RETRIEVAL-AUGMENTED GENERATION (RAG) FOR TIME SERIES

Although Large Language Models (LLMs) have achieved remarkable success in natural language tasks such as summarization, question answering, and reasoning, they struggle in time-series domains to reason over temporal structure or draw on prior sensor context. Most notably, they cannot dynamically access external data and are prone to hallucinations when faced with unfamiliar input [9].

Retrieval-Augmented Generation (RAG) was introduced to mitigate these limitations. In RAG, external retrieval modules first fetch relevant context from a knowledge base, which is then fused with the input query and passed to a language model for response generation. The architecture typically consists of two modules (illustrated in Fig. 3):

- **Retriever:** This module indexes an external corpus—often represented as high-dimensional embeddings—and performs similarity-based search against user queries. Common systems include FAISS and Milvus [29], [30], [32].
- **Generator:** The LLM consumes both the query and the retrieved context to produce an answer that is grounded, informative, and context-aware.

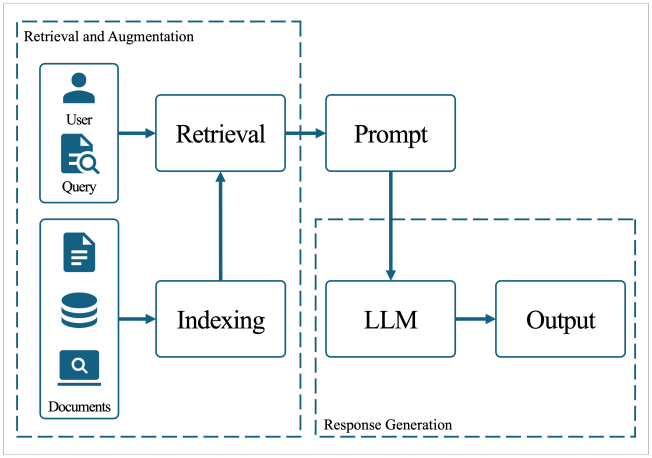


FIGURE 3. General architecture of retrieval-augmented generation (RAG). An external retriever indexes a document corpus as vector embeddings and returns relevant context at query time; the retrieved evidence is composed with the user query into a prompt that is passed to a large language model (LLM) to produce a grounded, context-aware response.

Unlike existing time series RAG approaches which target forecasting or clustering tasks, we apply RAG for interpretable anomaly classification — a new application area. For instance, [10] and [11] employ similarity-based time series retrieval to enhance forecasting, while [31] combines clustering and DTW-based search to support explanatory generation for multivariate data [33].

However, these prior works are largely designed for prediction tasks. In contrast, we propose a novel application of RAG for *interpretable anomaly detection*.

We extend the standard RAG pipeline with temporally-aware retrieval and statistical evidence construction tailored

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for anomaly decisions. Our method first segments the time series and embeds each segment using a deep encoder (T-Rep) [21]. The embeddings are queried against a vector database to retrieve historical segments with similar behavior. These retrieved sequences—annotated with their known anomaly status—are used to construct a prompt containing:

- Target segment metadata,
- Similar segments with distances and labels,
- Vector similarity scores (e.g., Cosine similarity, L2 Distance).

This enriched prompt enables the LLM to make both a binary anomaly decision and generate a reasoned explanation. This not only improves robustness by grounding inference in prior data but also enhances interpretability—making the system more transparent to human operators.

III. BACKGROUND

Because the proposed framework relies on retrieving similar historical patterns to guide LLM-based explanations, the embedding space must capture both structural and temporal context. As the backbone of the retrieval module, TS2Vec is adopted and extended with a time-aware embedding mechanism (T-Rep) [20], [21], [22]. This section describes the core learning objectives of [20], [21], formalized as four loss functions.

A. TEMPORAL-LEVEL CONTRASTIVE LOSS

To capture local continuity, temporal consistency is enforced within overlapping windows of the same series. Let $r_{i,t}$ be the representation of time t from sequence i, and $r'_{i,t}$ its augmented counterpart. The temporal loss encourages their similarity while penalizing alignment with negative pairs from the same window. Here, Ω denotes the index set of overlapping time steps used to sample augmented pairs.

$$\mathcal{L}_{\text{temp}}^{(i,t)} = -\log \frac{e^{r_{i,t}\cdot r'_{i,t}}}{\sum_{t'\in\Omega}\left(e^{r_{i,t}\cdot {r'}_{i,t'}} + \mathbb{I}_{[t\neq t']}e^{r_{i,t}\cdot r_{i,t'}}\right)} \tag{1}$$

Preserving local continuity helps the encoder distinguish natural temporal variation from structurally abnormal fluctuations.

B. INSTANCE-LEVEL CONTRASTIVE LOSS

To enforce inter-series discrimination, instances are contrasted across the batch. For each time step t in sequence i, the representation $r_{i,t}$ is encouraged to stay close to its augmentation $r'_{i,t}$ while remaining distant from representations of other sequences $j \neq i$ within the same batch. The denominator includes negative samples drawn across different sequences at the same time step:

$$\mathcal{L}_{\text{inst}}^{(i,t)} = -\log \frac{e^{r_{i,t}\cdot r'_{i,t}}}{e^{r_{i,t}\cdot {r'}_{j,t}} + \mathbb{I}_{[i\neq j]}e^{r_{i,t}\cdot r_{j,t}}} \tag{2}$$

This helps the model learn dataset-wide generalizable features, ensuring that retrieved similar segments truly represent comparable behavior.

C. DUAL CONTRASTIVE OBJECTIVE

The dual contrastive objective combines (1), (2) across all samples in a batch of N sequences, each with T time steps:

$$\mathcal{L}_{\text{dual}} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\mathcal{L}_{\text{temp}}^{(i,t)} + \mathcal{L}_{\text{inst}}^{(i,t)}\right) \tag{3}$$

This combined loss ensures that each representation is both temporally stable and discriminative across series—key properties for high-quality anomaly retrieval.

D. DIVERGENCE-AWARE TEMPORAL EMBEDDING

To inject position-awareness, T-Rep augments each time step t with a temporal embedding τ_t via Time2Vec. Building on [21], the embedding space is regularized by regressing the Jensen–Shannon divergence (JSD) between time embeddings τ_t and $\tau_{t'}$ against the distance between representation vectors $z_{i,t}, z'_{j,t'}$:

$$\mathcal{L}_{\text{div}} = \frac{1}{M}\sum_{(i,j,t,t')\in\Omega}^{M}\left(\mathcal{G}_1\left(z_{i,t} - z'_{j,t'}\right) - \text{JSD}(\tau_t \parallel \tau_{t'})\right)^2 \tag{4}$$

where $\mathcal{G}_1$ is a two-layer MLP with ReLU activations. The objective is to align latent distances with temporal divergence in embedding space. By aligning latent differences with temporal divergence, (4) encourages embeddings that encode both pattern and position.

E. TIME-CONDITIONED REPRESENTATION FORECASTING

To further strengthen temporal understanding and robustness to missing data, [21] introduce a prediction loss that regresses the representation at a future step $t + \Delta_t$ conditioned on current state $z_{i,t}^{(c_1)}$ and the corresponding time embedding $\tau_{t+\Delta_t}$:

$$\mathcal{L}_{\text{pred}} = \frac{1}{MT}\sum_{j\in\Omega_N}\sum_{t\in\Omega_T}\left(\mathcal{G}_2\left(\left[z_{i,t}^{(c_1)}; \tau_{t+\Delta_t}\right]^T\right) - z_{i,t+\Delta_t}^{(c_2)}\right)^2 \tag{5}$$

$\mathcal{G}_2$ is a multi-layer perceptron and c_1, c_2 denote two stochastic augmentations of the same sequence. The objective in (5) encourages structure-preserving, time-aware embeddings. Forecasting representations under random augmentation effectively simulates missing or delayed readings, improving robustness to incomplete sequences.

Taken together, these objectives yield embeddings that are discriminative, temporally aware, and resilient to noise—properties required by the downstream retrieval-augmented explanation pipeline.

IV. METHODOLOGY

This section details a framework for explainable time series anomaly detection comprising five stages: segmentation, deep feature–based retrieval, spatial comparison, query augmentation, and LLM-based decision generation. An overview is provided in Fig. 4.

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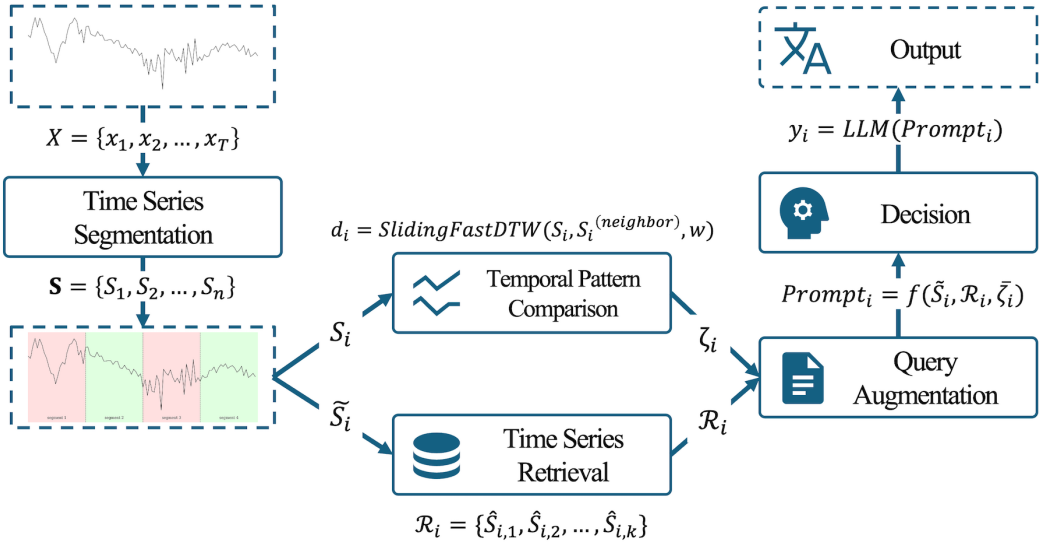


FIGURE 4. **Overview of the proposed framework for explainable time series anomaly detection. A univariate series X is segmented into candidate subsequences $S = \{S_i\}$; for each segment S_i, a context window is appended and the segment is embedded by a time-aware encoder. The system retrieves the top-k structurally similar historical segments $\mathcal{R}_i$ from a vector database and computes a Z-score–based anomaly index ζ_i by aligning S_i with time-aligned neighbor sensors via a sliding-window FastDTW procedure. A structured prompt Prompt_i then summarizes metadata, similarities, and ζ_i; a large language model (LLM) outputs a binary decision y_i together with a concise natural-language rationale.**

A. TIME SERIES SEGMENTATION

Let $X = \{x_t\}_{t=1}^{T}$ be a univariate time series. We aim to partition X into a set of non-overlapping segments $\mathbf{S} = \{S_i\}_{i=1}^{n}$, where each segment S_i is a candidate for anomaly inspection. Accurate segmentation allows the model to isolate semantically meaningful sequences, improving the granularity of anomaly reasoning. We use 1D U-Net for its ability to capture both local and global structures efficiently.

B. DEEP FEATURE-BASED TIME SERIES RETRIEVAL

For contextualization, each segment S_i is extended into $\tilde{S}_i$ by prepending a fixed-length window of preceding observations:

$$\tilde{S}_i = \left\{x_{t_i-\tau},\ x_{t_i-(\tau-1)},\ \ldots,\ x_{t_i}\right\} \tag{6}$$

where τ is the context window and t_i the start index of S_i. The extended segment $\tilde{S}_i$ is embedded with a T-Rep encoder, and Top-K similar sequences $\mathcal{R}_i = \{\hat{S}_{i,k}\}_{k=1}^{K}$ are retrieved from a vector database. Retrieving structurally similar historical segments supplies reference patterns under comparable conditions, helping to separate rare-but-valid deviations from true anomalies.

To improve robustness against outliers, mean squared error is replaced by the Huber loss [34]. MSE can overweight large errors caused by anomalies and destabilize the embedding space; the Huber loss limits this influence and yields more stable representations for retrieval:

$$\text{Huber}(r) = \begin{cases} \frac{1}{2}r^2 & \text{if } |r| \le \delta \\ \delta(|r| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \tag{7}$$

The Huber-enhanced divergence and prediction losses are given by:

$$\mathcal{L}_{div}^{\text{Huber}} = \frac{1}{M} \sum_{(i,j,t,t')\in\mathcal{P}}^{M} \text{Huber}\left(\mathcal{G}_1(z_{i,t}, z'_{j,t'}) - \text{JSD}(\tau_t \,\|\, \tau_{t'})\right) \tag{8}$$

$$\mathcal{L}_{\text{pred}}^{\text{Huber}} = \frac{1}{MT} \sum_{j\in\Omega_N}^{M} \sum_{t\in\Omega_T}^{T} \text{Huber}\left(\mathcal{G}_2\left(\left[z_{i,t}^{(c_1)}; \tau_{t+\Delta_j}\right]^T\right) - z_{i,t+\Delta_j}^{(c_2)}\right) \tag{9}$$

These modifications enhance training stability in the presence of anomalous or noisy segments.

C. SPATIAL NEIGHBOR COMPARISON VIA FASTDTW

Algorithm 1 *Sliding Window FastDTW*

Input: Target segment S_i, Neighboring segment $S_i^{(neighbor)}$, window size w
Output: Distance score d_i

```
procedure SlidingFastDTW(S_i, S_i^(neighbor), w)
    Initialize empty list D
    compare_window ← min(|S_i|, |S_i^(neighbor)|)
    for t = 0 to compare_window − w step w
        x ← S_i[t : t + w]
        y ← S_i^(neighbor)[t : t + w]
        d_t ← FastDTW(x, y)
        Append d_t to D
    end for
    d_i ← mean(D)
end procedure
```


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To differentiate local anomalies from contextually valid deviations, each segment S_i is compared with time-aligned segments from spatially adjacent sensors $S_i^{(neighbor)}$ using FastDTW algorithm [33]. A sliding window is applied to compute the average distance

Algorithm 2 *Statistical Computation of DTW Distance Distributions*

Input: Time series data from all sensor pairs
Output: Mean μ_d and standard deviation σ_d of DTW distances

procedure ComputeDTWStats()
  window_size $\leftarrow 6$
  stride_size $\leftarrow 3$
  for each sensor pair (s_1, s_2) **do**
    for each measurement e **do**
      Extract time series T_1, T_2 for (s_1, s_2) and e
      Calculate missing ratio
      if missing ratio > 0.1 **then**
        Skip this segment
      else
        Replace missing with 0
      end if
      Slice both series using *window_size* and *stride_size*
      for each sliced pair (x, y) **do**
        Compute DTW distance $d_i \leftarrow \text{FastDTW}(x, y)$
        Append d_i to D
      end for
    end for
  end for
  $\mu_d \leftarrow \frac{1}{|D|} \sum d_i$
  $\sigma_d \leftarrow \sqrt{\frac{1}{|D|} \sum (d_i - \mu_d)^2}$
return μ_d, σ_d
end procedure

To disentangle localized deviations from broader contextual shifts, each candidate segment S_i is contrasted with time-aligned segments from spatially adjacent sensors $S_i^{(neighbor)}$ using a sliding-window DTW procedure with the FastDTW approximation (Algorithm 1) [33]. The resulting mean distance d_i is standardized into a Z-score anomaly index ζ_i (11) using precomputed normal-period statistics (Algorithm 2). This spatial corroboration down-weights segments that co-vary with neighbors and elevates uniquely deviating segments, improving the reliability of anomaly decisions.

$$d_i = \text{SlidingFastDTW}(S_i, S_i^{(neighbor)}, w) \tag{10}$$

This distance is standardized into a Z-score using reference statistics μ_d and σ_d so that scores are comparable across sensors with different baselines and can be thresholded consistently:

$$\zeta_i = \frac{d_i - \mu_d}{\sigma_d} \tag{11}$$

The parameters (μ_d, σ_d) are estimated from historical inter-sensor DTW distances as described in Algorithm 2.

When a segment S_i has multiple spatial neighbors, let $\zeta_i = \left\{\zeta_i^{(1)}, \ldots, \zeta_i^{(m)}\right\}$ denote the set of Z-scores computed via FastDTW against each neighbor. These values are aggregated into a single scalar $\bar{\zeta}_i$ as the mean absolute Z-score over $m = |\mathcal{N}_i|$ neighbors:

$$\bar{\zeta}_i \leftarrow \frac{1}{|\mathcal{N}_i|} \sum_{j=1}^{|\mathcal{N}_i|} |\zeta_i^{(j)}| \tag{12}$$

D. QUERY AUGMENTATION FOR PROMPT CONSTRUCTION

Prompts are constructed by fusing retrieved evidence and statistical signals so the LLM can reason over both data-driven patterns and contextual cues. The anomaly score ζ_i, retrieved similar sequences $\mathcal{R}_i$, and segment metadata are combined to generate a prompt $Prompt_i$ that feeds into the LLM. This prompt encapsulates contextual evidence, including weighted anomaly probabilities.

Prompt construction logic is detailed in Algorithm 3.

Algorithm 3 *Prompt Generation via Query Augmentation*

Input: Metadata of segment $\tilde{S}_i$, Retrieved similar time series $\mathcal{R}_i = \{\hat{S}_{i,k}\}_{k=1}^{K}$, mean neighbor Z-score $\bar{\zeta}_i$
Output: Natural language prompt for LLM: $Prompt_i$

procedure $f(\tilde{S}_i, \mathcal{R}_i, \bar{\zeta}_i)$
  Initialize *abnormal_weight* $\leftarrow 0$, *normal_weight* $\leftarrow 0$, *total_weight* $\leftarrow 0$
  min_dist $\leftarrow \infty$
  for each $(\hat{S}_{i,k}, cos_dist_{i,k}) \in \mathcal{R}_i$ **do**
    $w \leftarrow 1/(cos_dist_{i,k} + \epsilon)$
    if $CLASS(\hat{S}_{i,k}) = 1$ **then** // Abnormal data
      abnormal_weight $\leftarrow$ *abnormal_weight* $+ w$
    else if $CLASS(\hat{S}_{i,k}) = 0$ **then** // Normal data
      normal_weight $\leftarrow$ *normal_weight* $+ w$
    end if
    total_weight $\leftarrow$ *total_weight* $+ w$
    min_dist $\leftarrow \min(min_dist,\ d)$
  end for
  if *min_dist* $= \infty$ **or** *min_dist* $> \delta$ **then**
    Final_abn $\leftarrow 100$
    // Mark as unseen if no sufficiently similar match
    // (cosine threshold δ)
  else
    $P_{abn} \leftarrow \frac{abnormal_weight}{total_weight} \times 100$
    Final_abn $\leftarrow 0.5 \times P_{abn} + 0.5 \times (1 - \bar{\zeta}_i) \times 100$
    // k denotes the number of top similar sequences retrieved (Top-k)
  end if
  Construct natural language $Prompt_i$ using $\tilde{S}_i$, $\mathcal{R}_i$, ξ_i, *Final_abn*, and metadata
end procedure

The resulting prompt, formatted in markdown for clarity and compatibility with LLMs, is shown in Appendix Fig. 8.

E. NATURAL LANGUAGE GENERATION AND DECISION

Finally, the constructed prompt $Prompt_i$ is passed into the LLM function f_{LLM} to produce an output $y_i = \{decision_i, e_i\}$, where $decision_i \in \{0, 1\}$ is a binary decision indicating whether the input segment is anomalous, and e_i is a natural language explanation justifying that decision.

$$y_i = \text{LLM}(Prompt_i) \tag{13}$$

This design enables the model to not only classify each segment, but also to provide interpretable rationales grounded in retrieved evidence and spatial context.

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V. EXPERIMENTAL RESULTS AND COMPARISONS

A. DATASET

Experiments use real-world data from the South Korean National Air Quality Monitoring Network. The corpus comprises hourly measurements from 2021–2024 across more than 600 monitoring stations distributed nationwide. Each station reports five key pollutant concentrations: **CO, NO, NO_2, O_3, SO_2**. Expert annotators provide anomaly labels for a subset of segments To separately assess the encoder and the end-to-end framework, the data are partitioned into three disjoint subsets: (1) an encoder pretraining set for unsupervised representation learning, (2) a test set for measuring embedding generalization, and (3) a pipeline evaluation set that simulates deployment conditions with unseen data.

- **Encoder Pretraining Set (2021–2023):** Used to train the time series encoder (T-Rep) in a self-supervised manner. Resulting embeddings are stored in a vector database for retrieval.
- **Encoder Test Set (Jan–Apr 2024):** Used to evaluate the generalization of the pretrained encoder on unseen data by measuring retrieval precision and embedding quality.
- **Pipeline Evaluation Set (Jul–Aug 2024):** Used to assess the full anomaly detection pipeline, including segmentation, retrieval, spatial scoring, and LLM-based explanation. This data is strictly held out from training and retrieval setup.

Only approximately **0.5%** of all hourly data points in the dataset are labeled as anomalous. This reflects the rarity of critical pollution events or sensor faults in stable environmental monitoring systems, and poses a significant challenge for accurate and class-balanced detection.

As illustrated in Fig. 5, Labeled anomalies vary substantially in duration, ranging from brief events under 24 hours to persistent deviations lasting over six months. Most anomaly segments fall between 1 day and 2 weeks in length. This diversity in anomaly duration poses a challenge for fixed-length segmentation and motivates our use of flexible, context-aware detection methods.

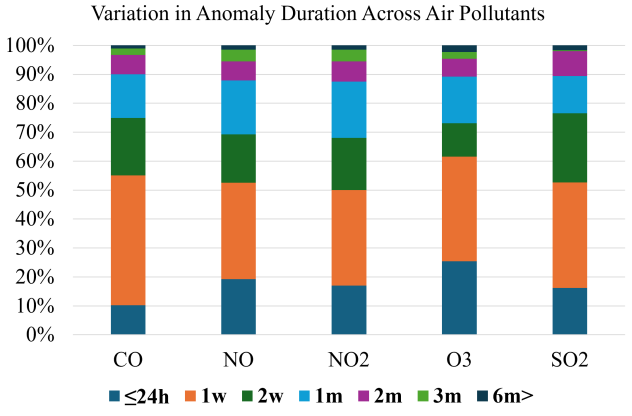


FIGURE 5. Anomaly duration by pollutant. Stacked bars report, for each pollutant, the within-pollutant percentage of segments in duration bins ($<$24 h, 1 w, 2 w, 1–2–3 m, $\geq$6 m).

Fig. 6 shows hourly CO concentration from July 2024 at a selected station. Although no explicit anomalies were annotated, missing values are present. Fig. 7 displays NO concentrations from August 2024, with a confirmed anomalous segment highlighted.

B. EXPERIMENTAL SETUP AND EVALUATION METRICS

The system is evaluated using standard binary classification metrics: **Recall**, **Precision**, **F1-Score**, and **Accuracy**.

Two phases of evaluation were conducted:

- **Phase 1: Encoder Quality Evaluation.** After self-supervised training of the encoder (e.g., T-Rep), representations for labeled segments were used to train a Support Vector Machine (SVM) [39]. The classification accuracy of this SVM reflects the encoder's capability to capture meaningful representations suitable for anomaly-related downstream tasks.
- **Phase 2: End-to-End Framework Evaluation.** The complete pipeline — including segmentation, retrieval, spatial comparison, and natural language reasoning — was tested against baseline models. We primarily compare with a segmentation-only system that detects anomalies based solely on thresholded output, without retrieval or language-based interpretation.

C. RESULTS ON REPRESENTATION QUALITY EVALUATION

To compare different encoder configurations, we evaluate three variants: **TS2Vec**, **T-Rep (MSE)**, and **T-Rep (Huber)**. TS2Vec is implemented using its original design, producing 320-dimensional latent vectors. T-Rep is evaluated under two loss settings—MSE and Huber—using 128-dimensional embeddings, consistent with its original architecture focused on memory efficiency and temporal precision.

Table 1 to Table 2 show the classification performance using the SVM classifier trained on these representations. While TS2Vec benefits from a higher-dimensional representation (320 dimension) and strong contrastive pretraining, T-Rep demonstrates superior anomaly-specific feature encoding despite its lower dimensionality (128 dimension), especially when trained with Huber loss. These results affirm the utility of Huber loss in real-world noisy time series settings and demonstrate the value of T-Rep (Huber) as a foundation for retrieval-based anomaly detection.

D. RESULTS ON OVERALL FRAMEWORK EVALUATION

The proposed anomaly detection framework was evaluated on real-world air quality data collected in South Korea from July to August 2024. The dataset contains hourly measurements for five pollutants (CO, NO, NO_2, O_3, SO_2), with 108 abnormal samples annotated by domain experts.

Table 3 reports detection counts only. The 1D U-Net predicts 2,322 anomalous segments with 89 true positives (TPs), whereas the proposed framework predicts 1,094 segments with 88 TPs [35]. This corresponds to a reduction of false positives from 2,233 to 1,006 (**55%** fewer) at comparable sensitivity.

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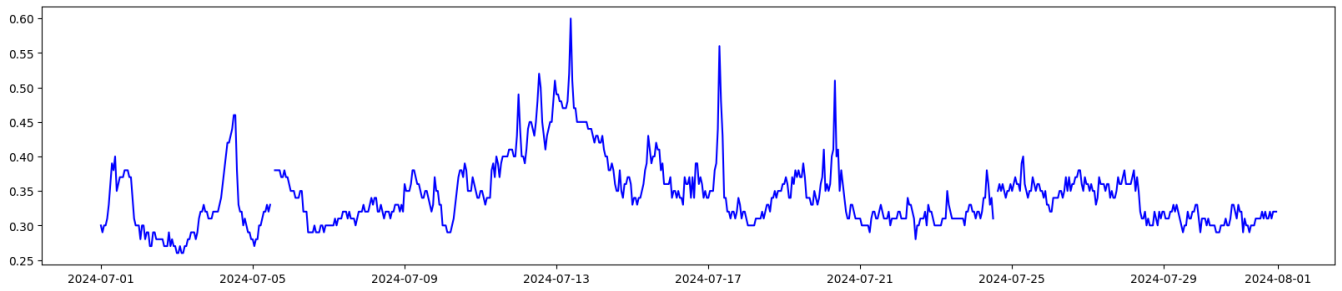


FIGURE 6. **Hourly CO concentration at a monitoring station during July 2024. No expert-labeled anomalies occur in this period.**

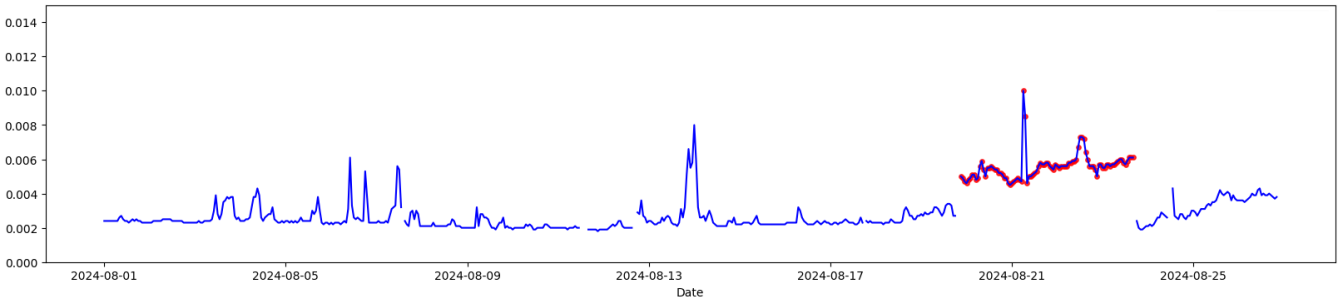


FIGURE 7. **Hourly NO concentration at a monitoring station during August 2024. The red segment marks an expert-labeled anomaly, and the remaining blue portion shows non-anomalous behavior.**

TABLE 1. **Anomaly classification performance of encoder variants across pollutants. The best-performing values for each pollutant are highlighted in blue.**

Element	Normal/Abn	TS2Vec				T-Rep (MSE)				T-Rep (Huber)			
		Recall	Prec.	F1	Acc.	Recall	Prec.	F1	Acc.	Recall	Prec.	F1	Acc.
CO	565/52	0.57	0.69	0.62	0.94	0.56	0.79	0.51	0.95	**0.76**	**0.91**	**0.83**	**0.97**
NO	253/13	0.57	0.69	0.62	**0.97**	0.42	0.68	0.52	0.96	**0.71**	**0.71**	**0.71**	**0.97**
NO_2	273/13	0.58	**0.68**	0.61	**0.97**	0.47	0.53	0.48	0.95	**0.74**	0.65	**0.69**	**0.97**
O_3	152/12	0.40	0.61	0.47	0.94	0.06	0.13	0.08	0.92	**0.87**	**0.87**	**0.87**	**0.98**
SO_2	374/27	0.70	0.78	0.74	0.97	0.31	0.64	0.40	0.94	**0.84**	**0.94**	**0.89**	**0.99**

TABLE 2. **Overall classification performance across encoder configurations. The best-performing values for each metric are highlighted in blue.**

Model	Normal/Abn	Recall	Prec.	F1	Acc.
TS2Vec	1617/117	0.56	0.76	0.65	0.96
T-Rep (*MSE*)		0.37	0.55	0.43	0.94
T-Rep (*Huber*)		**0.78**	**0.82**	**0.79**	**0.98**

TABLE 3. **Detection counts on real-world monitoring data (July–August 2024). The proposed framework reduces false positives by 54.9% (2322→1094) while keeping true positives nearly identical (89 vs. 88).**

Element	# Abn (Samples)	1D U-Net		Proposed	
		# Detected	# True Pos.	# Detected	# True Pos.
CO	41	1007	33	441	32
NO	22	231	20	108	20
NO_2	22	317	20	213	20
O_3	3	207	2	122	2
SO_2	20	560	14	210	14
Total	108	2322	**89**	**1094**	88

Tables 4 and 5 summarize accuracy and Recall, respectively, for the baseline, the proposed method, and six supervised sequence classifiers [15], [19], [23], [36], [37], [38]. The proposed method attains the best overall accuracy (**0.93**) and matches the 1D U-Net in total recall (**0.84**). Supervised classifiers show noticeably lower recalls in this imbalanced, real-world setting, underscoring the robustness of the retrieval-augmented decision process compared with direct segment-level classification.

In summary, the proposed framework delivers a stronger precision–recall trade-off for operational deployment by reducing false positives while maintaining detection effectiveness across pollutant types. This makes the system especially suitable for regulatory or alerting applications, where minimizing false alarms is critical for operational trust.

VI. CONCLUSION

This work presented a multi-stage anomaly detection framework that couples deep time series representation learning with retrieval-augmented reasoning via large language models. The proposed approach was validated using real-world air

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TABLE 4. Accuracy comparison on real-world air-quality segments. Row-best values are shown in blue. The proposed method achieves the highest total accuracy (0.93) and is best on CO/NO/NO_2/O_3, while SO_2 shows ties among several deep baselines.

Element	Proposed	1D U-Net	Transformer	LSTM-FCN	TCN	InceptionTime	ResNet-1D	LSTM
CO	**0.93**	0.86	0.92	0.89	0.90	0.89	0.89	0.90
NO	**0.96**	0.91	0.85	0.92	0.91	0.92	0.89	0.92
NO_2	**0.93**	0.90	0.89	0.89	0.92	0.91	0.89	0.90
O_3	**0.96**	0.94	0.90	0.91	0.88	0.32	0.88	0.88
SO_2	0.89	0.84	**0.93**	**0.93**	**0.93**	0.87	0.92	0.92
Total	**0.93**	0.89	0.90	0.91	0.91	0.78	0.89	0.90

TABLE 5. Recall comparison on real-world air-quality segments. Row-best values are shown in blue. Total recall is tied between the proposed method and 1D U-Net (0.84); InceptionTime peaks on O_3, and the proposed method leads on SO_2.

Element	Proposed	1D U-Net	Transformer	LSTM-FCN	TCN	InceptionTime	ResNet-1D	LSTM
CO	0.78	**0.80**	0.35	0.07	0.30	0.16	0.68	0.19
NO	**0.90**	**0.90**	0.26	0.23	0.21	0.05	0.33	0.13
NO_2	**0.90**	**0.90**	0.55	0.26	0.33	0.10	0.45	0.10
O_3	0.76	0.76	0.17	0.83	0.10	**0.97**	0.00	0.00
SO_2	**0.83**	0.81	0.49	0.36	0.47	0.51	0.36	0.20
Total	**0.84**	**0.84**	0.39	0.22	0.31	0.21	0.48	0.16

quality monitoring data collected from nationwide stations in South Korea.

The proposed framework addresses the limitations of traditional segmentation-based anomaly detection methods, which tend to rely solely on abrupt changes in time series data, often resulting in high false positive rates. To mitigate this, the proposed framework comprises five sequential modules: (1) time series segmentation using a 1D U-Net, (2) deep retrieval using the T-Rep encoder, (3) spatial pattern comparison with neighboring sensor segments, (4) Z-score and vector similarity-based query augmentation, and (5) final decision generation with natural language explanations using LLMs.

Experimental results demonstrate that the T-Rep encoder trained with Huber loss consistently outperforms TS2Vec and MSE-based T-Rep models across all evaluation metrics. Notably, the Huber-based encoder achieved superior performance in terms of both precision and recall, validating its robustness against outliers and noisy data. Furthermore, the framework reduces total alarms by 52.9% and false positives by 54.9% compared to the 1D U-Net baseline [35], while maintaining comparable recall. This highlights the effectiveness of our retrieval-based conditional reasoning and prompt augmentation pipeline.

Overall, the framework achieves high precision and recall while delivering interpretable, context-aware explanations via large language models, which supports deployment in real-world monitoring and alerting environments.

Future work includes extending the framework to multivariate time series, assessing performance on complex anomaly types, improving the consistency and diversity of LLM-generated explanations, and enhancing the scalability of vector retrieval operations.

Findings are based on univariate hourly series from a single national network; generalization to other regions, sampling rates, or multivariate settings is not guaranteed. End-to-end performance remains sensitive to the initial segmentation stage because missed segments cannot be recovered downstream. Retrieval quality is bounded by encoder capacity and corpus coverage and may degrade under distribution shift. Although explanations are fluent, faithfulness to evidence is not fully assured without additional controls. Finally, large-scale embedding maintenance, DTW-based spatial scoring, and LLM prompting introduce compute and latency overheads; a comprehensive scalability analysis is left for future work.

APPENDIX A
IMPLEMENTATION DETAILS: PROMPT DESIGN AND INTERACTIVE INTERFACES

This appendix details practical aspects of the implementation to support reproducibility and user-centered deployment of the proposed framework. We describe the prompt design used for LLM-based anomaly interpretation and present two interactive interfaces that assist domain experts in reviewing and validating anomaly decisions.

A. PROMPT TEMPLATE FOR LLM-BASED ANOMALY REASONING

The prompt is constructed dynamically from metadata, retrieval results, statistical comparisons, and model outputs, as specified in Algorithm 3. A typical template includes the following:

- **Segment Metadata:** station/location, time span, and pollutant type (e.g., CO, NO).
- **Retrieved Similar Cases:** top-k historical segments with labels and distances.
- **Statistical Signal:** Z-score of spatial deviation.

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- **Estimated Abnormality:** anomaly probability weighted by retrieval similarity and statistical context.

Fig. 8 shows a representative example of the prompt submitted to the LLM.

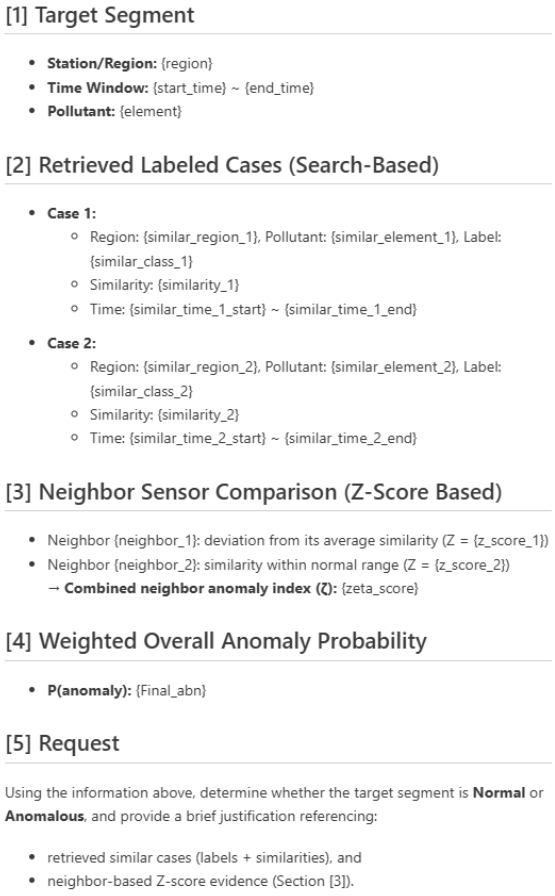


FIGURE 8. Generated prompt (English-localized) used for LLM-based anomaly decision and explanation. The template aggregates segment metadata (station, pollutant, time), top-k retrieved historical cases with labels and distances, a spatial Z-score, and a weighted abnormality estimate; together these fields ground the LLM's decision and rationale

The prompt enables the LLM to generate both a *Positive/Negative* decision and a narrative explanation grounded in historical similarity and statistical reasoning.

B. INTERACTIVE INTERFACES FOR HUMAN-IN-THE-LOOP VALIDATION

To improve interpretability and facilitate expert review, we implemented two user interfaces: a graph-based visualization and a chat-based interaction.

1) Graph-based Visualization Interface

Fig. 9 shows the web-based UI designed for expert anomaly inspection. When a user selects a target time segment on the plot, the system displays (i) the original time series with marked segments, (ii) retrieved similar sequences from historical data, (iii) associated labels and similarity/distance scores, and (iv) an LLM-generated explanation.

This interface is optimized for domain experts who prefer a visual dashboard to cross-reference multiple signals when validating anomaly decisions.

2) Chat-based Anomaly Explanation

To enhance accessibility for non-expert users, we developed a chatbot-style front end. As shown in Fig. 10, users can query a specific time span or segment, and the system returns an LLM-generated explanation that includes (i) a summary of the target segment, (ii) the relevance and outcomes of retrieved cases, (iii) statistical deviation signals such as the Z-score, and (iv) a final natural-language decision with its rationale.

This conversational interface fosters interpretability and encourages end-user trust in anomaly detection outputs, especially in sensitive domains such as environmental monitoring and facility operations.

Note: The interface is localized for NIER and originally displays Korean text; for publication, on-screen text is provided in English translation.

C. LLM OUTPUT FORMAT AND POST-PROCESSING

To ensure consistency and downstream utility, LLM outputs are post-processed into structured JSON records with the following fields: `segment_id`, `anomaly_decision`, `confidence_score`, and `explanation_text`. These structured outputs can be logged, visualized, or consumed by automated alerting systems. This modularity supports real-world deployment, scalability, and auditability of the proposed framework.

D. LLM CONFIGURATION AND TRAINING DETAILS

To implement natural language–based anomaly explanation, a publicly available Korean-language instruction-tuned model, **LLaMA-3 Korean Bllossom 8B**, provided by MLP-KTLim on Hugging Face[1] [28], was adopted. The model is based on Meta's LLaMA 3 8B backbone and fine-tuned to support high-quality Korean reasoning and instruction-following tasks [8]. This choice balances performance and computational efficiency and offers an open-access license suitable for academic use.

Prompt inputs were sized to remain within the 8B variant's context-window limits (8,192 tokens).

E. ENCODER TRAINING CONFIGURATION

For the deep feature–based time series encoder, the original T-Rep implementation was used as described in its reference paper. All hyperparameters, data-augmentation techniques, and loss weightings were kept identical to the official open-source release. To ensure robustness and reduce stochastic variation, training was repeated **30 times per experimental condition**, and metrics are reported as averages across these trials.

[1]https://huggingface.co/MLP-KTLim/llama-3-Korean-Bllossom-8B

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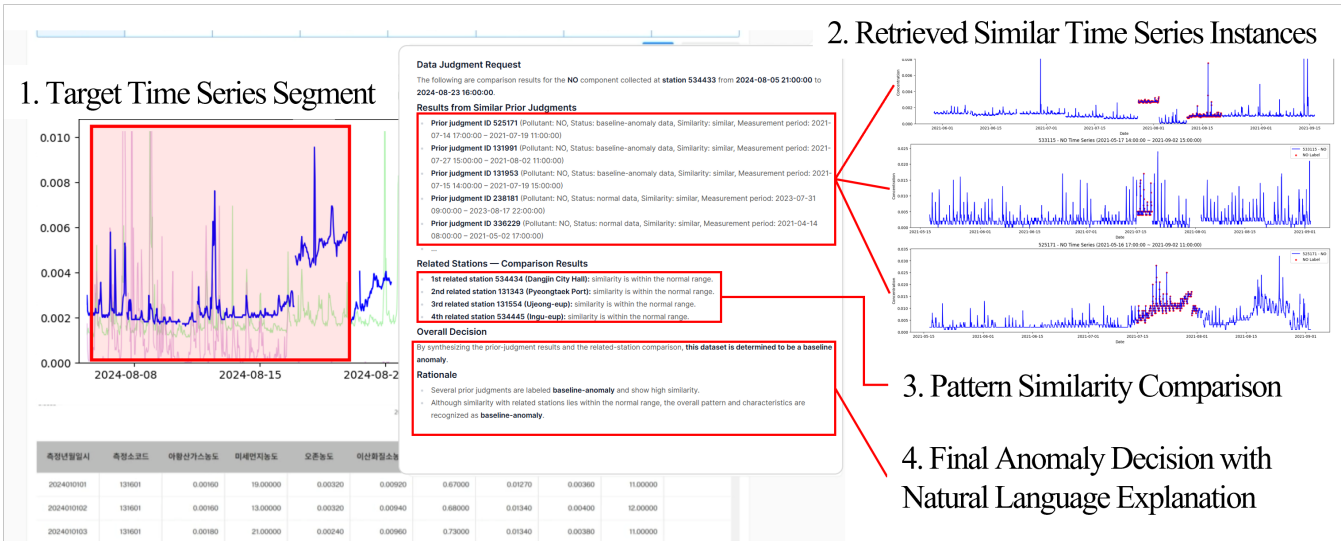


FIGURE 9. **Graph-based interface enabling visual inspection and explanation of detected anomalies.**

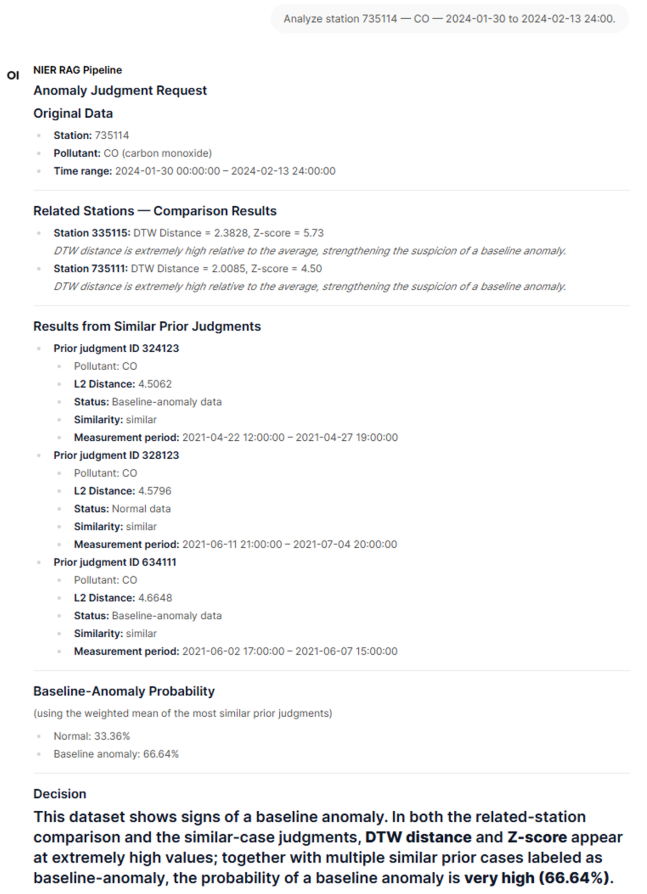


FIGURE 10. **Chat-based interface offering a conversational explanation of anomaly decision.**

By holding the encoder setup fixed and introducing novel components only in the downstream pipeline, improvements in end-to-end performance can be attributed to the proposed framework rather than encoder-specific tuning. This design helps bridge the gap between black-box anomaly detectors and explainable, human-interpretable reasoning—an essential requirement for real-world adoption in sensitive domains.

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